

Summarizing the Performance of Variable Spending Rules in the Historical Data

Thus far, we have compared the historical performance of various spending strategies when the initial spending rate is 4 percent. We have not yet applied a PAY Rule or considered how spending may be impacted by the low interest rate environment facing retirees. But before getting to that, it is worthwhile to introduce an exhibit that summarizes the information about the distribution of spending and wealth outcomes.

Exhibit 6.16 compares 4 percent initial spending rates for each of the spending rules using the historical data that we have just analyzed visually. This exhibit allows us to see how spending evolves over time at different percentiles of the distribution, as well as remaining wealth, direction of spending over time, and the average spending reductions over time. It also summarizes and quantifies the previous visuals about how spending and wealth evolved in each of the rolling historical periods.

For each strategy, the initial spending rate is shown, with the assumption that retirement wealth is equal to \$100,000. Results are scalable for other wealth amounts. The distribution for real spending amounts is shown for ten, twenty, and thirty years into retirement. The distribution of remaining real wealth is shown after the thirtieth year of retirement. Considering spending and wealth are both important—as retirees should not be narrowly focused on a singular goal to avoid financial wealth depletion—financial goals for retirement can essentially be reduced to two competing objectives: to support as much spending as feasible, and to maintain a reserve of financial assets to support risk management objectives such as protecting from spending shocks or otherwise provide a legacy.

Three additional spending metrics are included at the right of the exhibit.

1. The **change in real spending after thirty years** for that strategy relative to its initial spending level. Positive numbers mean that spending increases over retirement, while negative numbers imply